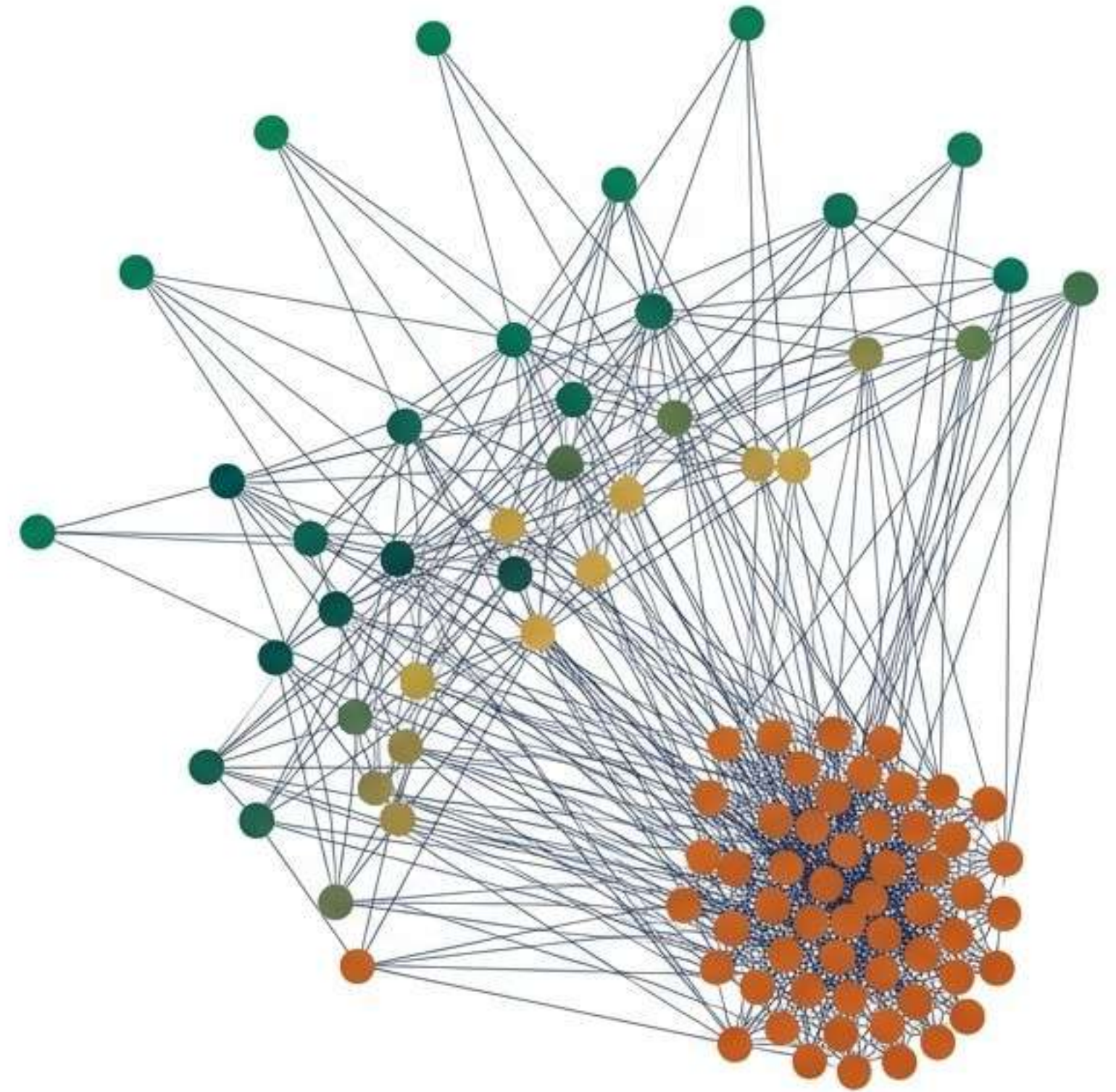


Credit Risk Profiling & Feature Feature Discovery

Uncovering the 'DNA' of default
through behavioral data analysis.



Executive Summary: The Profile of High Risk



The 'Bad' Definition

Risk is defined as >30 Days Past Due (DPD). This creates a 5% vs 95% class imbalance, requiring robust sampling techniques.



Demographic Drivers

Youth is a risk. Customers aged 25-30 show the highest default density. Counter-intuitively, 'Married' applicants show higher default rates than single ones.



Economic Signals

"Pegawai Swasta" (Private Sector) employees drive the majority of defaults. Entrepreneurs and Civil Servants showed zero defaults in this sample.

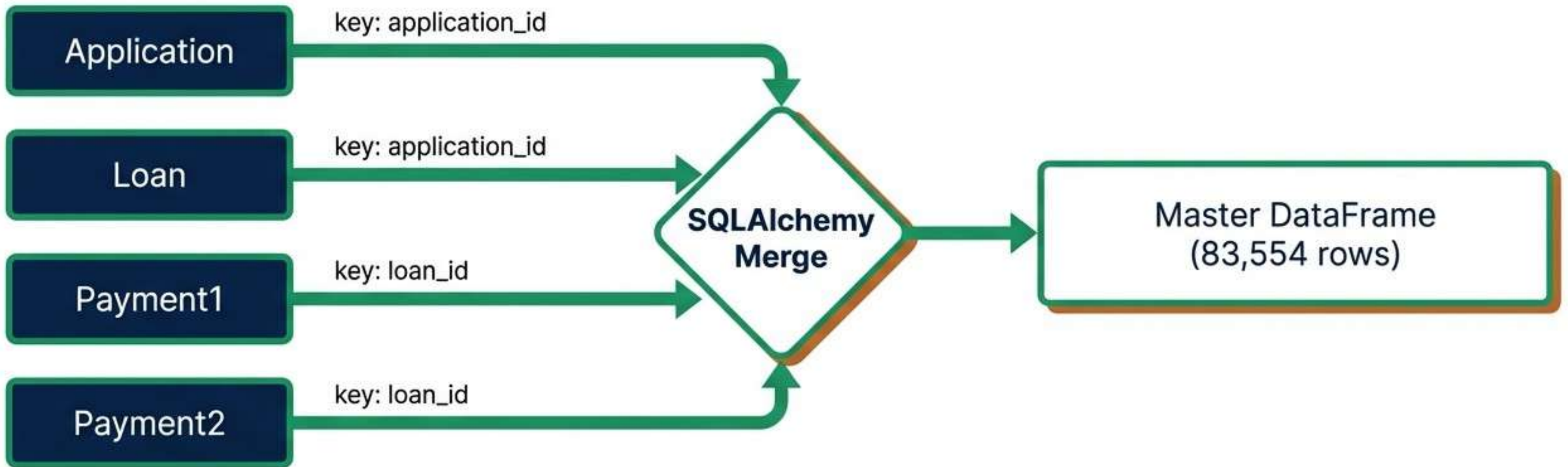


Product Risk

Loan duration is a friction point. Longer loans correlate with higher default rates. Features like "Outstanding Balance" are critical predictors.

Data Architecture: Establishing the Single Source of Truth

Consolidating fragmented sources into a master analytical record.



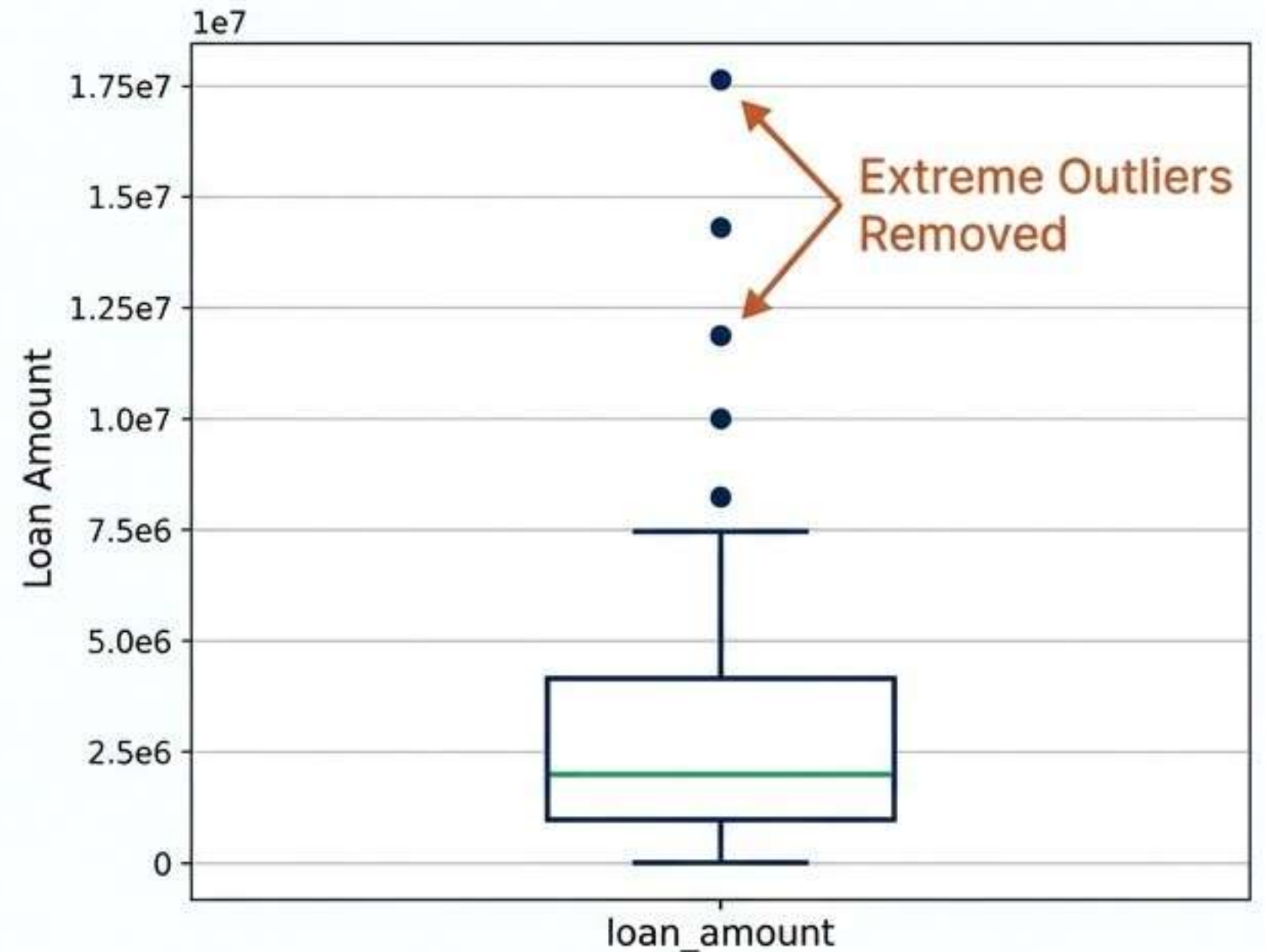
```
merge_df = query_from_db('SELECT a.*, b.*, c.*, d.* FROM application a LEFT JOIN loan b  
ON a.app_id = b.app_id LEFT JOIN payment1 c ON...')
```


Data Hygiene: Noise Reduction & Outlier Treatment

Improving model stability by removing extreme anomalies.

Methodology: Interquartile Range (IQR)

- **Loan Amount:** Removed 26 extreme outliers (Values < -3.25M or > 6.75M).
- **Installment Amount:** Removed 23 outliers.
- **DPD Normalization:** Removed negative values (early payments) to focus on delinquency.

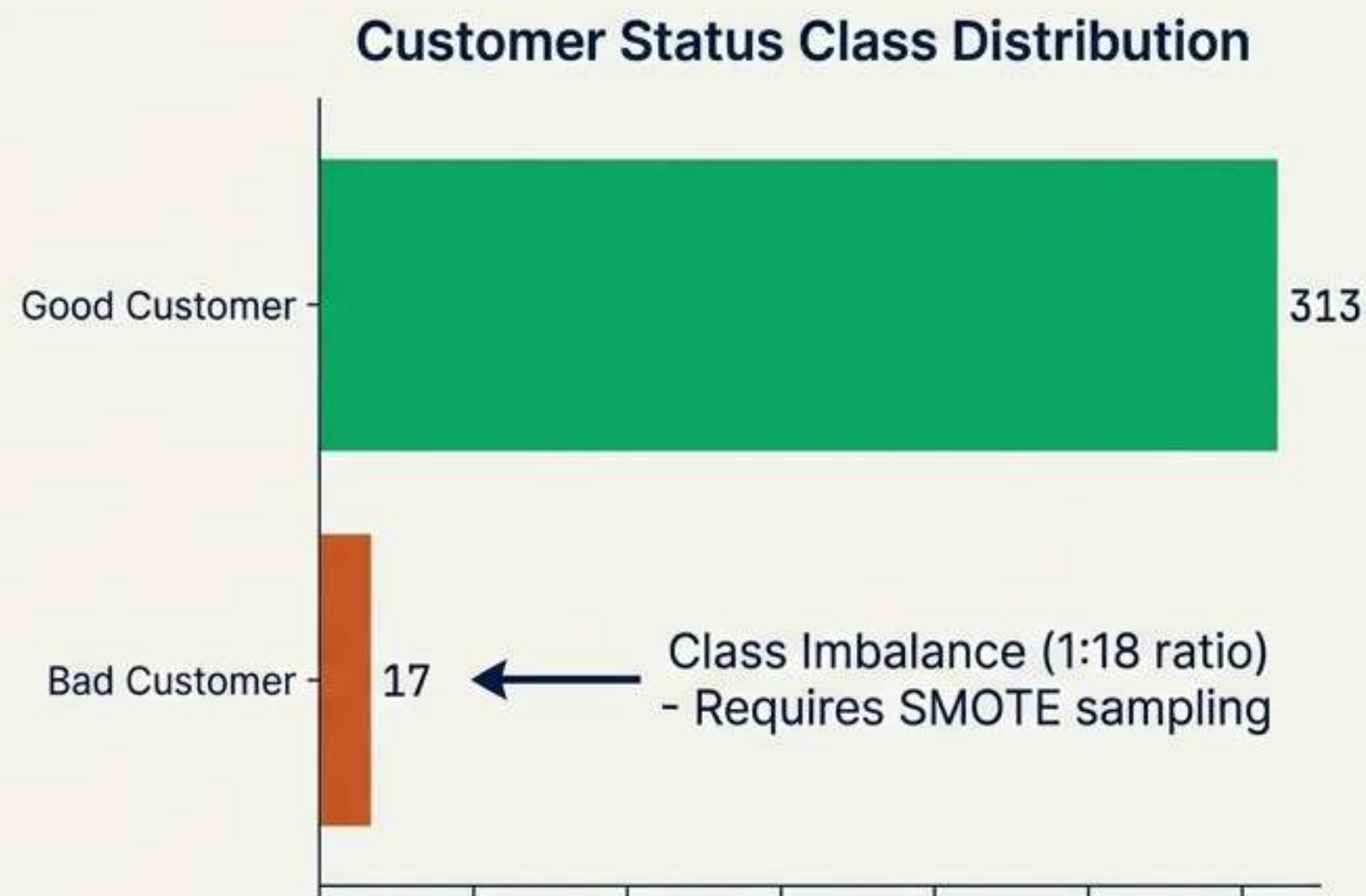


Defining the 'Bad' Customer

Deriving ground truth labels from repayment behavior.

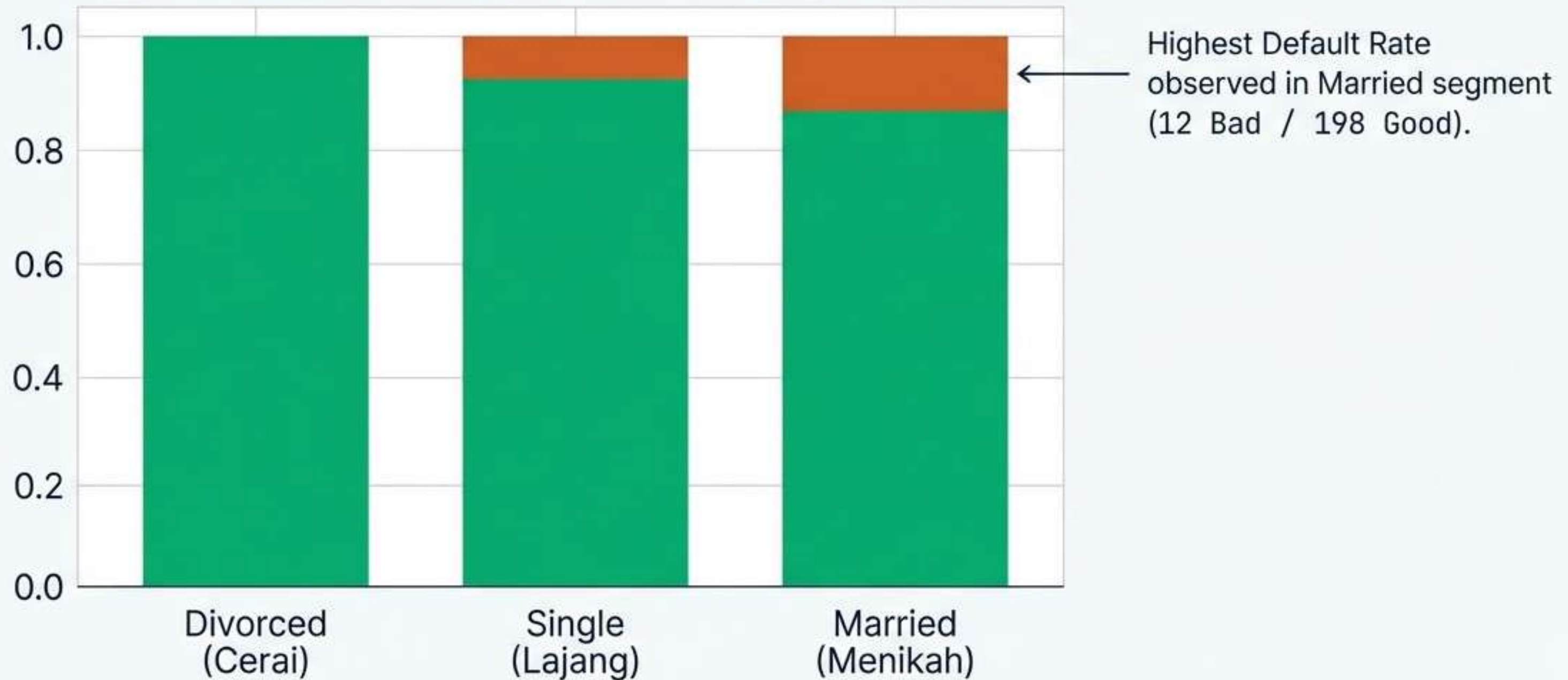
DPD > 30 Days
= Bad Customer

DPD ≤ 30 Days
= Good Customer



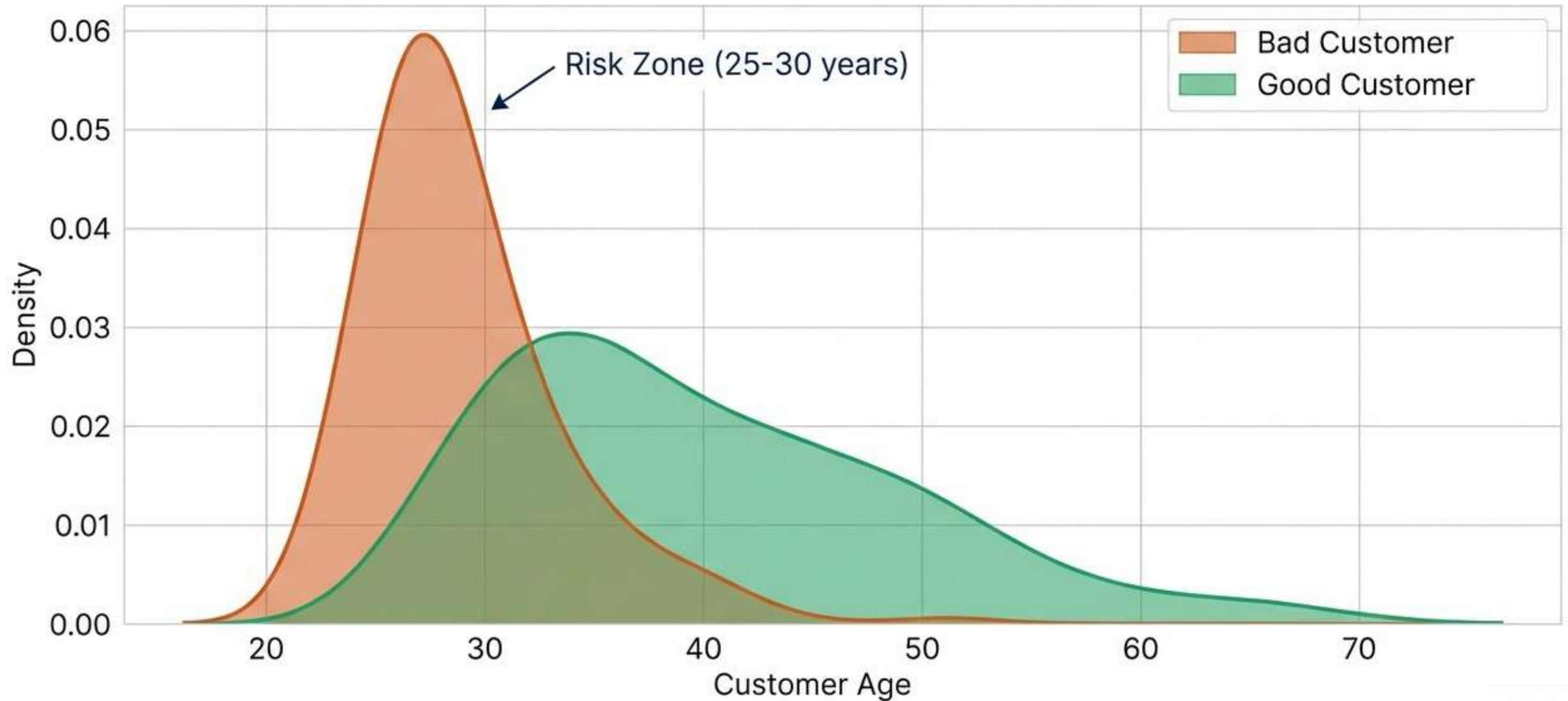
Risk Profile: Marital Status

Counter-intuitive insight regarding applicant stability.



Risk Profile: The Impact of Age

Younger borrowers represent the highest concentration of default risk.



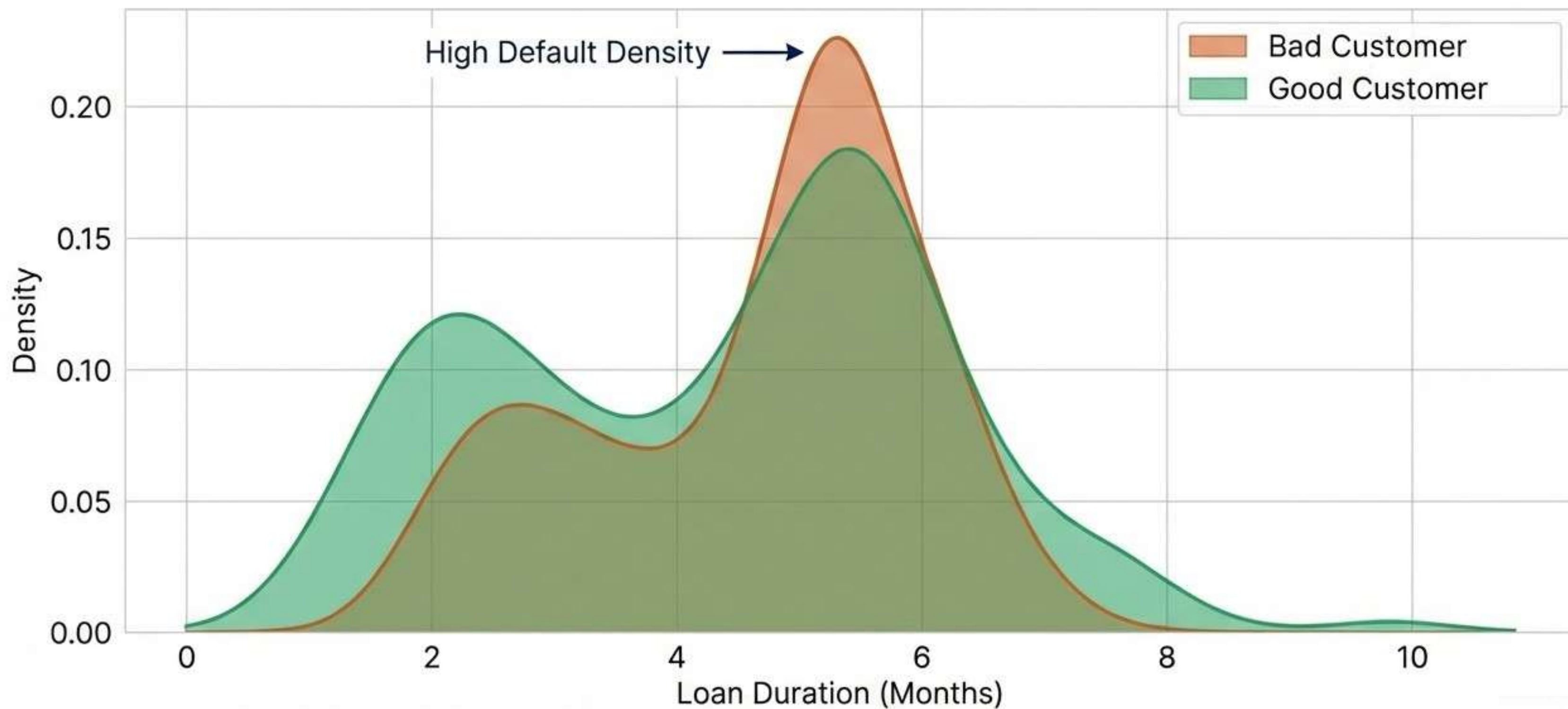
Risk Profile: Employment & Industry

Private sector employees drive the volume of risk.



Risk Profile: Loan Characteristics

Loan duration acts as a friction point for repayment.



Risk Stratification: Decile Analysis

Evaluating risk distribution across loan magnitudes.

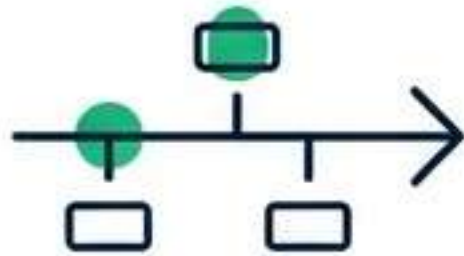
Decile	No. Loans	No. Defaulted	Default Rate	Cumulative Rate
1	X,XXX	XXX	X.XX%	X.XX%
2	X,XXX	XXX	X.XX%	X.XX%
3	X,XXX	XXX	X.XX%	X.XX%
4	X,XXX	XXX	X.XX%	X.XX%
5	X,XXX	XXX	X.XX%	X.XX%
6	X,XXX	XXX	X.XX%	X.XX%
7	X,XXX	XXX	X.XX%	X.XX%
8	X,XXX	XXX	X.XX%	X.XX%
9	X,XXX	XXX	X.XX%	X.XX%
10	X,XXX	XXX	X.XX%	X.XX%

Methodology: SQL NTILE(10) function used to partition customers by Loan Amount to test if higher loans carry higher proportional risk.

Feature Engineering: Transforming Data into Signals

Creating predictive variables for the modeling phase.

Feature: Customer Age



Current Date - Date of Birth = Age

```
df['age'] = (now - df['dob']) / 365
```

Why:

- Validates the 'Youth Risk' hypothesis observed in EDA.

Feature: Outstanding Balance



Loan Amount - Total Paid Amount = Balance

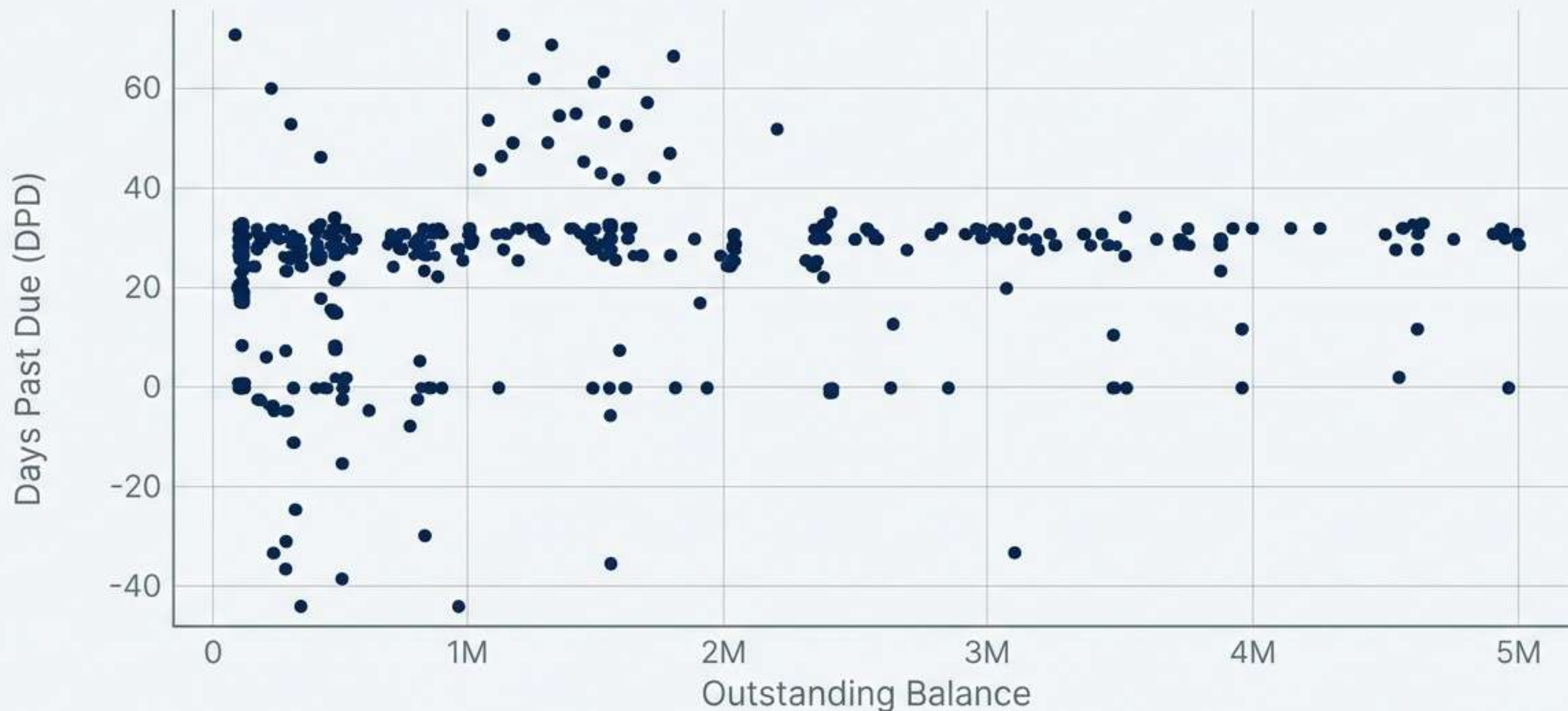
```
df['balance'] = df['loan'] - df['paid_p1']
```

Why:

- High remaining balance relative to tenure signals distress.

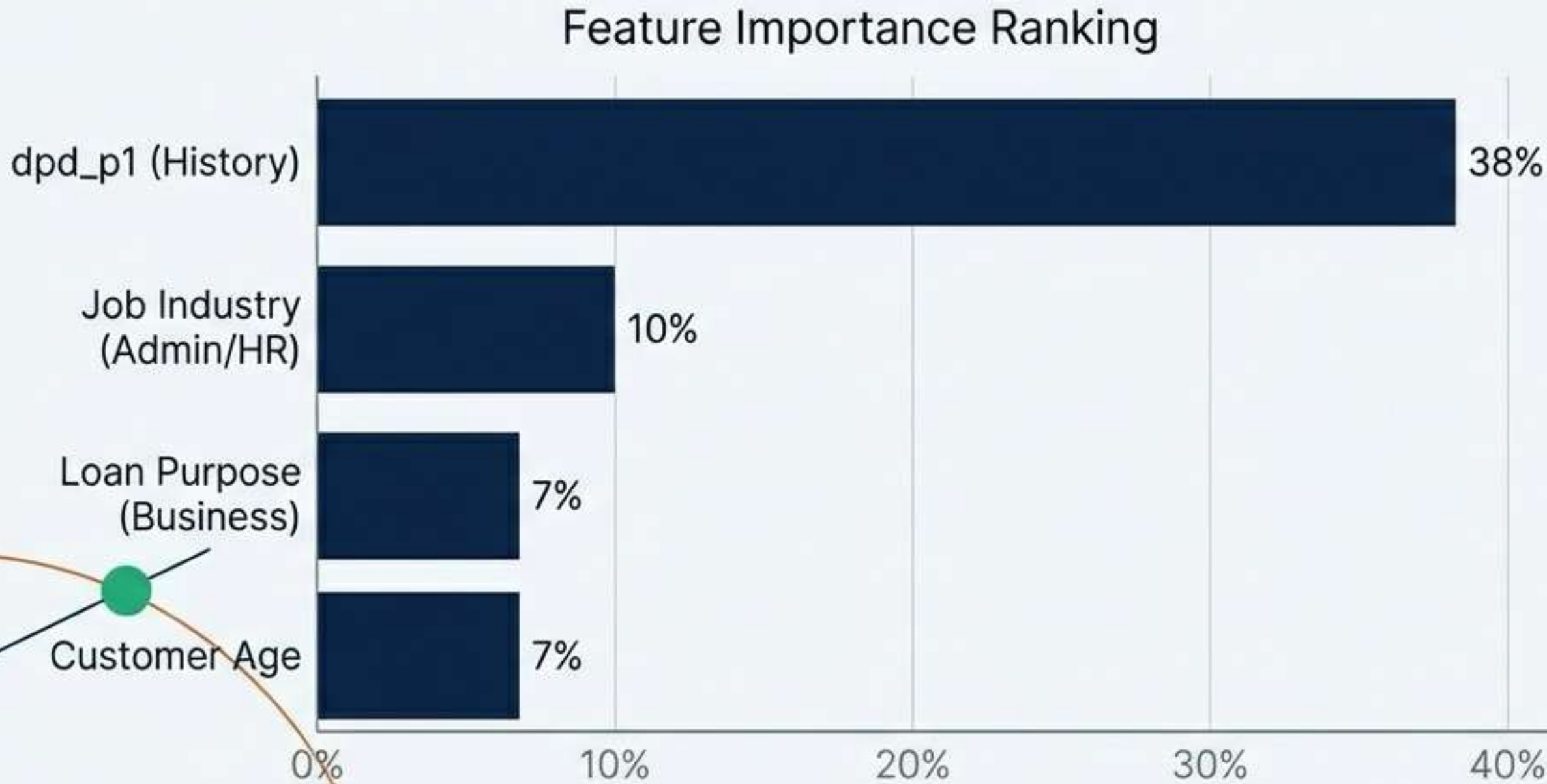
Feature Validation: Outstanding Balance vs. Risk

Visual confirmation of the engineered feature's predictive power.



The DNA of a Defaulter: Feature Importance

Top predictors identified by the Random Forest classifier.



- Behavioral history (DPD) is the dominant predictor, followed by socio-economic factors (Job/Industry). This confirms the 'DNA' profile constructed during analysis.

Strategic Recommendations

Translating data science findings into business rules.



Targeting

Implement stricter credit scoring for applicants under 30. Review approval thresholds for 'Private Sector' employees.



Product

Limit exposure on 5-6 month loan durations. Monitor 'Outstanding Balance' > **1.5M IDR** as an early warning signal.



Model Readiness

Engineered features (Age, Balance, DPD) are validated. Proceed to training with **SMOTE sampling** to handle 95/5 imbalance.